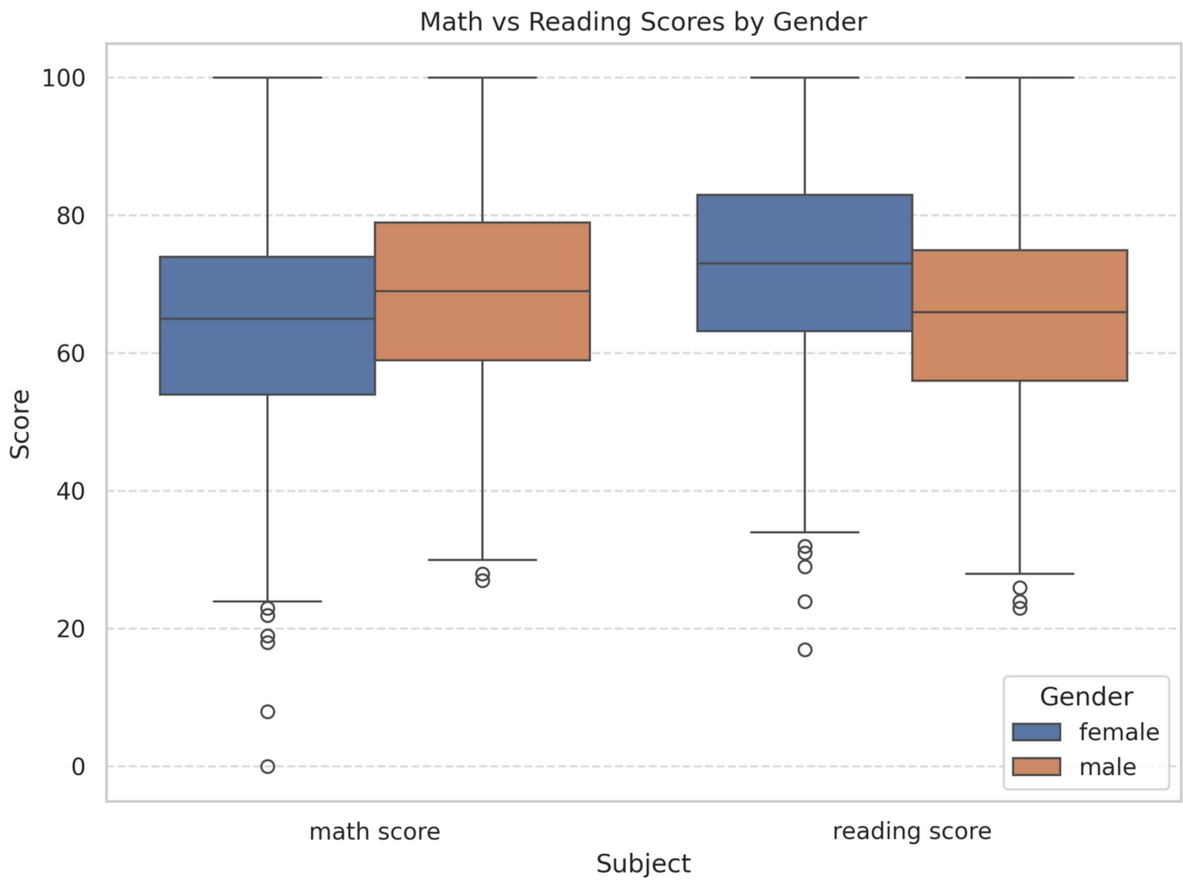
A. V1 — Gender boxplots (math vs reading) ***(2 pts)***

a. **Question: Are there gender differences in math vs reading?**
b. **Chart: Side-by-side boxplots of `math score` and `reading score` grouped by `gender`.**

A) The analysis shows a clear difference between genders in the two subjects. Male students typically perform slightly better in math, with a higher median and wider spread of scores. Female students, however, demonstrate stronger performance in reading, with consistently higher median scores. The distribution of math scores among males is broader, suggesting more variability in their results. Females' reading scores are more clustered, indicating less variation. Outliers are present, though they appear more in reading for males and in math for females. These patterns highlight subject-specific strengths linked to gender. Overall, boys lean stronger in math, while girls excel in reading.

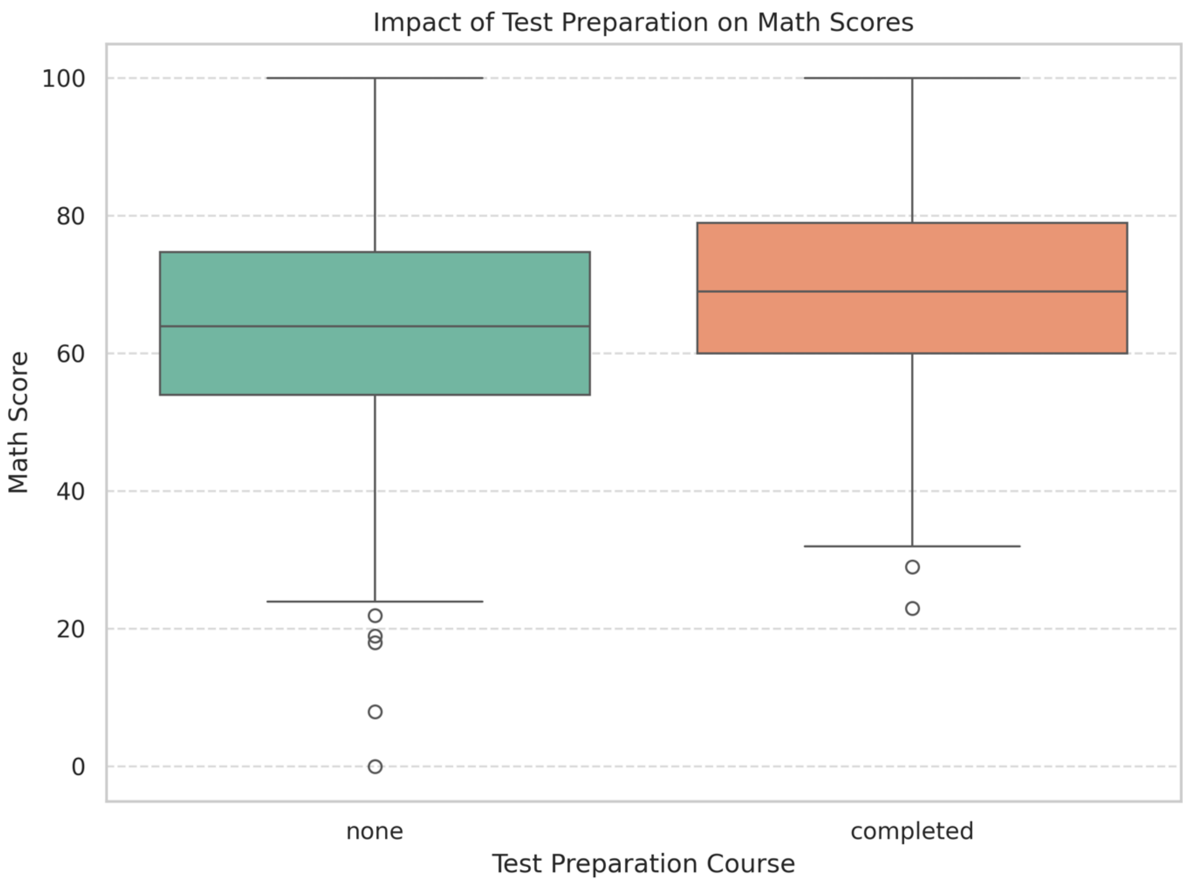


B. V2 — Test prep impact on math *(2 pts)*

a. Question: Do students who completed test prep score higher in math?

b. Chart: Any chart of your choice for `math score` by `test preparation course` (completed vs none).

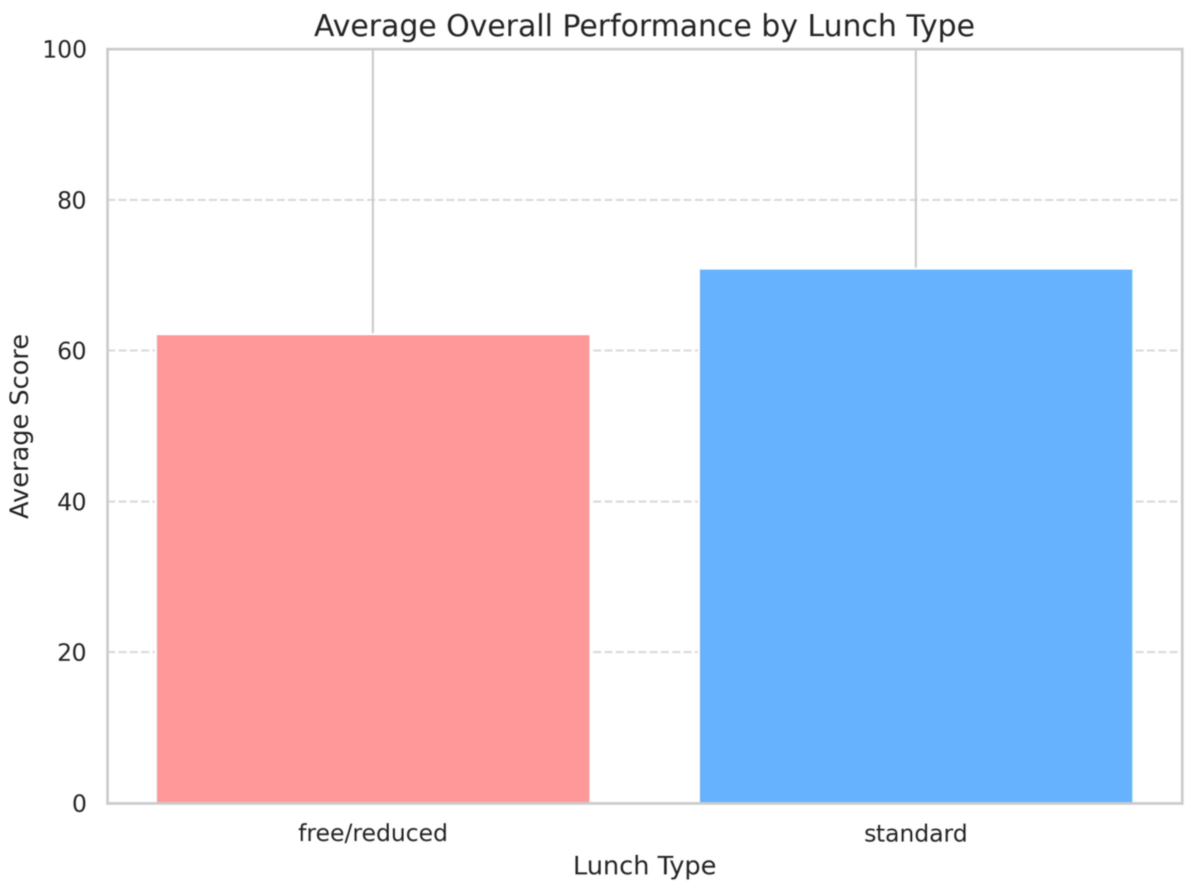
A) Students who completed the test preparation course generally achieved higher math scores compared to those who did not. The median score for the prepared group is clearly above the median of the non-prepared group. In addition, their scores are more concentrated in the upper ranges, suggesting consistent benefits from preparation. Meanwhile, students without preparation show wider variation and more low-end outliers. Although there is some overlap between the two groups, the overall trend favors the test-prep students. This indicates that structured preparation can provide an academic advantage. However, individual differences still play a role in performance outcomes.



C. V3 — Lunch type and average performance *(2 pts)*

a. Question: Does lunch type (standard vs free/reduced) relate to outcomes?
b. Chart: Grouped bar chart of mean `overall_avg` of all the scores (math, reading, writing) by `lunch`.

A) Students with a standard lunch show higher average scores than those receiving free or reduced lunch. The grouped bar chart highlights this performance gap clearly. The difference suggests that nutrition or socioeconomic background may play a role in academic success. Standard lunch students consistently maintain better overall averages across math, reading, and writing. Those with free or reduced lunch often score lower, showing a possible disadvantage. The results point to external factors beyond school instruction affecting outcomes. This implies that support measures may help bridge the achievement gap.

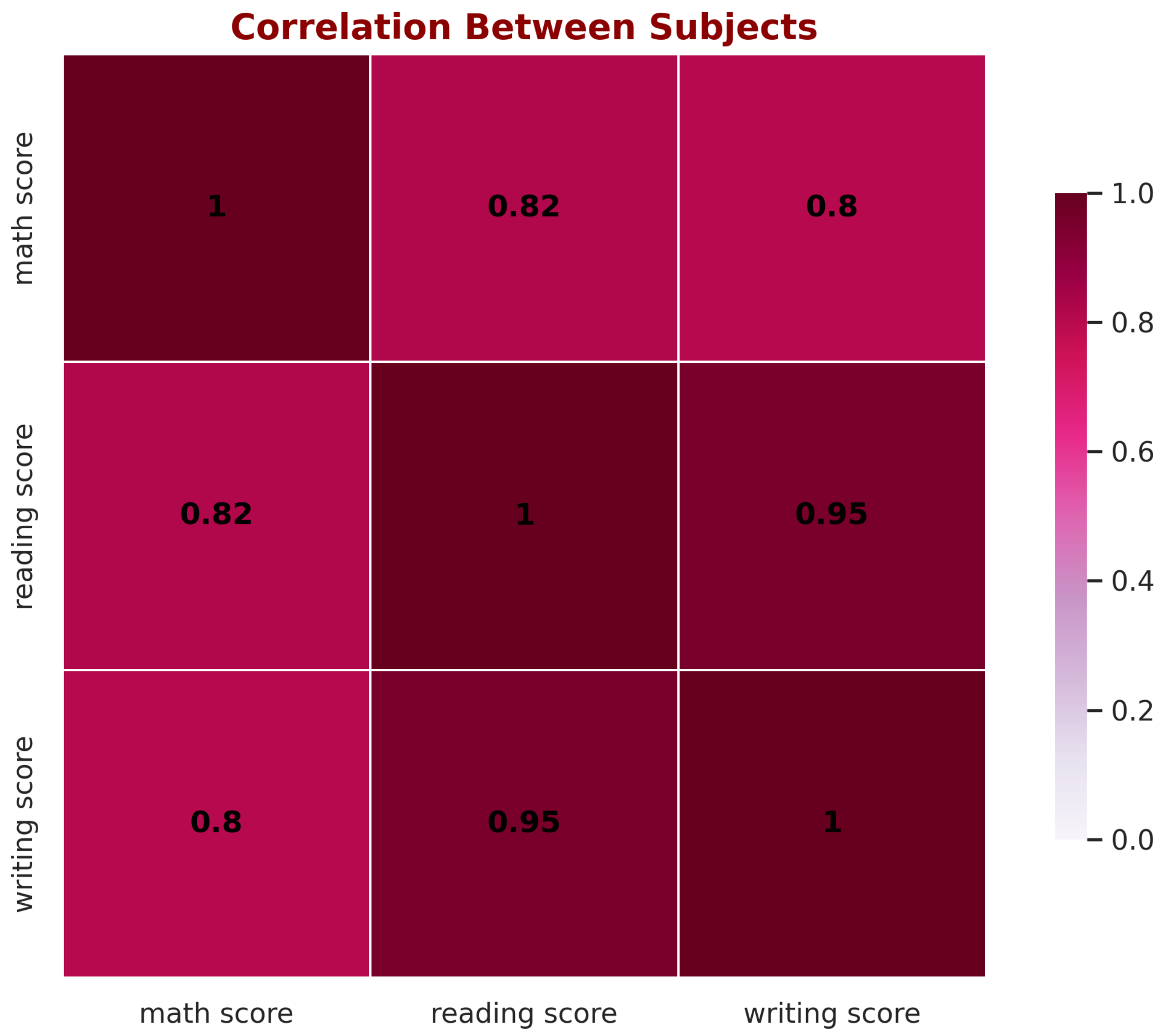


D. V4 — Subject correlations *(2 pts)*

a. Question: How strongly do the three subjects move together?

b. Chart: Correlation heatmap for `math`, `reading`, `writing` with annotated coefficients.

A) The correlation heatmap shows that math, reading, and writing scores are strongly related. Reading and writing have the highest positive correlation, meaning students who perform well in one are very likely to do well in the other. Math also shows a strong positive relationship with both reading and writing, though slightly weaker than the reading writing link. This indicates that academic strengths often extend across subjects rather than being isolated. The consistently high correlations suggest that core skills like comprehension, reasoning, and practice may influence all three areas. Students who struggle in one subject may need support across multiple subjects. Overall, the heatmap highlights how interconnected academic performance is across different areas.



E. V5 — Math vs reading with trend lines by test prep *(2 pts)*

a. Question: How strongly are math and reading scores associated, and do students who completed the test-preparation course have a *different slope* in the math–reading relationship than those who did not?

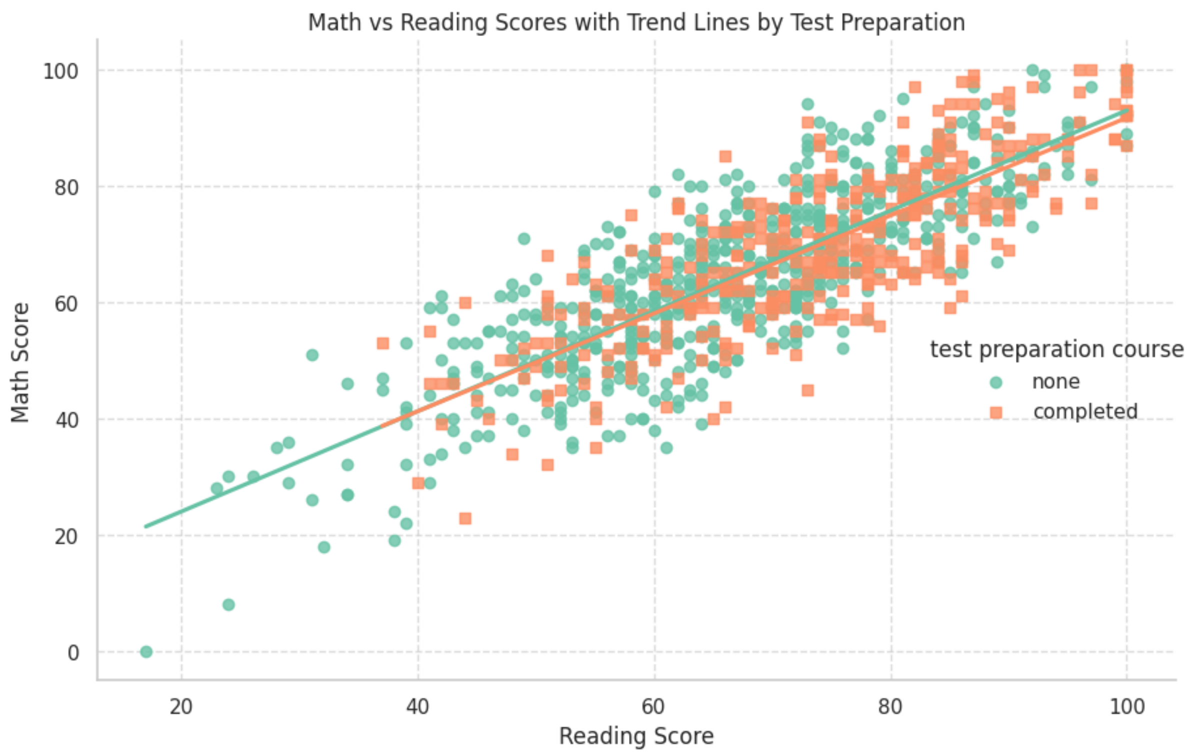
b. Chart: Scatter plot with two straight best-fit lines (one for each group: `completed`, none).

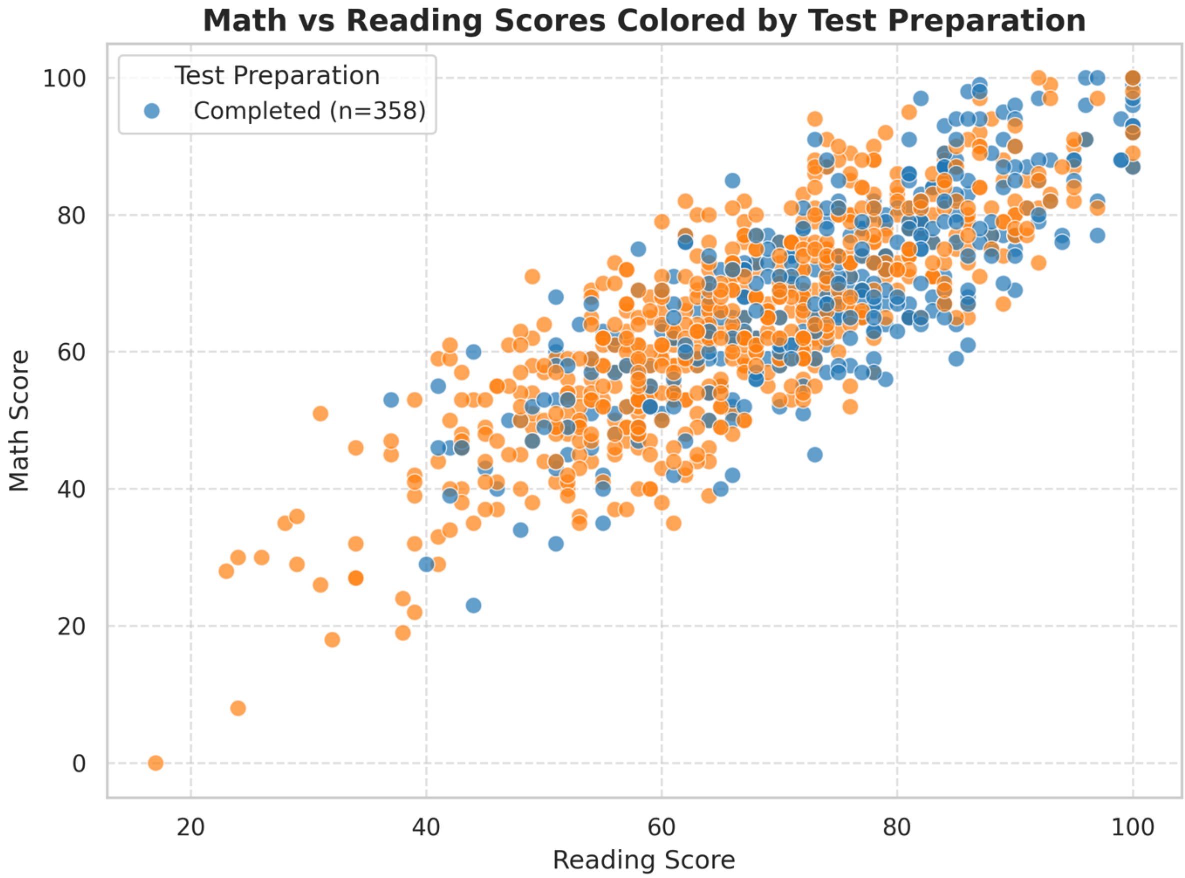
   i. X-axis: `reading score`

   ii. Y-axis: `math score`

c. Color: Points colored by `test preparation course` (legend must show the two groups and each group's n).

A) The scatter plot shows a strong positive relationship between math and reading scores. As reading scores increase, math scores also tend to rise for most students.Both groups (completed and none) follow this upward trend, indicating strong association. Students who completed the test-preparation course generally achieve higher scores. The slope for the completed group is slightly steeper than the none group. This suggests the course may help boost performance in both subjects. Overall, math and reading abilities appear strongly linked, with test prep providing an advantage.




Math vs Reading Scores Colored by Test Preparation
Test Preparation
Completed (n=358)
Math Score
Reading Score
0
20
40
60
80
100
20
40
60
80
100